复旦数学资料汇编
模形式

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第一章　模形式

1.1　作业: 模形式

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来自王海宁的手写习题 pdf.

In the following text, we guarantee that $\mathbb{H} = \{z \in \mathbb{C}, \mathrm{im}(z) > 0\}$, E_n be the weight-n normalized Eisenstein series, $\mathfrak{D}$ be the fundamental domain w.r.t. $\mathrm{SL}_2(\mathbb{Z})$.

1.2　第一次作业

1. Let $d\mu = \frac{dxdy}{y^2}$, show that $d\mu$ is a $\mathrm{SL}_2(\mathbb{Z})-$invariant measure on $\mathbb{H}$.

   证明 (解答)**.** *By complex analysis, we have $dxdy = 2idzd\bar{z}$, by the lesson we have*

   $$\frac{d(gz)d(g\bar{z})}{dzd\bar{z}} = \frac{d(gz)}{dz} \cdot \frac{d(g\bar{z})}{d\bar{z}} = \frac{1}{(cz+d)^2(c\bar{z}+d)^2} = \frac{1}{|cz+d|^4}.$$

   $$\frac{\mathrm{im}(gz)}{\mathrm{im}(z)} = \frac{1}{|c\bar{z}+d|^2}.$$

   Combine these two above, we have done.

2. Find the value of $\iint_{\mathfrak{D}} \frac{dxdy}{y^2}$.

   证明 (解答)**.**

   $$\iint_{\mathfrak{D}} \frac{dxdy}{y^2} = \int_{-\frac{1}{2}}^{\frac{1}{2}} dx \int_{\sqrt{1-x^2}}^{+\infty} \frac{dy}{y^2} \tag{1.1}$$

   $$= \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} \tag{1.2}$$

   $$= \frac{1\pi}{3}. \tag{1.3}$$

3. Show that the Bernoulli number vanishes for odd $k \geq 3$.

   证明 (解答)**.** *By the definition of Bernoulli numbers,*

   $$\frac{t}{e^t - 1} = \sum_{n=0}^{+\infty} B_n \frac{t^n}{n!}$$

So

$$\frac{t(e^t+1)}{2(e^t-1)} = \sum_{n\neq 1} B_n \frac{t^n}{n!},$$

because $\frac{t(e^t+1)}{2(e^t-1)}$ *is an even function, each of the* $B_k, k \neq 1,\ k$ *odd vanishes.*

4. Show that for even $k \geq 2$, $(-1)^{\frac{k}{2}} B_k < 0$.

证明 (解答)**.** *By the definition of Bernoulli Polynomials,*

$$\frac{te^{xt}}{e^t-1} = \sum_{n=0}^{+\infty} B_n(x)\frac{t^n}{n!}, \quad \textit{with}\ \ B_n(0) = B_n.$$

In particular, if we extend 1*-periodically of the* $[0,1)$ *value of Bernoulli Polynomials to the whole* $\mathbb{R}$*, we gain the periodic Bernoulli function* $\tilde{B_n}(x)$*, and its Fourier series are:*

$$\tilde{B_{2m}}(x) = (-1)^{m-1} \cdot \frac{2(2m)!}{(2\pi)^2 m} \sum_{n=1}^{+\infty} \frac{\cos(2\pi n x)}{n^{2m}}.$$

So push $x = 0$ *into it we gain:*

$$\begin{aligned} B_{2m} &= \tilde{B_{2m}}(0) && (1.4)\\ &= (-1)^{m-1}\frac{2(2m)!}{(2\pi)^{2m}} \sum_{n=1}^{+\infty} \frac{1}{n^{2m}} && (1.5)\\ &= (-1)^{m-1}\frac{2(2m)!}{(2\pi)^{2m}} \zeta(2m). && (1.6) \end{aligned}$$

and then we have done.

5. Let $\Delta = \frac{1}{1728}(E_4^3 - E_6^2)$. Show that the q-expansion of Δ has integral coefficients.

证明 (解答)**.** *Immediately from the formula* $\Delta = q\prod_{i=1}^{+\infty}(1-q^i)^{24}$.

6. Suppose f has integral q-expansion coefficients **with modularity**, show that f can be expressed by an **integral** polynomial in E_4, E_6 and Δ.

证明 (解答)**.** *We prove by induction on the weight* l *of the modular form* f *12-stepwisely.*
Because we have already known $\dim(M_{2k}(1)) = 1$*, for* $k = 2,3,4,5$*, and we have the following formulas:*

$$E_4 = 1 + 240\sum_{n=1}^{\infty}\sigma_3(n)q^n, \quad g_2 = (2\pi)^4\frac{1}{2^2\cdot 3}E_2$$

$$E_6 = 1 - 504\sum_{n=1}^{\infty}\sigma_5(n)q^n, \quad g_3 = (2\pi)^6\frac{1}{2^3\cdot 3^3}E_3$$

$$E_8 = 1 + 480\sum_{n=1}^{\infty}\sigma_7(n)q^n$$

$$E_{10} = 1 - 264\sum_{n=1}^{\infty}\sigma_9(n)q^n$$

$$E_4^2 = E_8,\ \ E_4 \cdot E_6 = E_{10}$$

we proved the conclusion for $k \leq 10$.

Then because $\times\Delta : M_k(1) \to S_{k+12}(1)$ *gives an isomorphism, we may write* $f = f_0 + m \cdot E_l$, *where* f_0 *is a cusp form and* $m \in \mathbb{Z}$, *then* $\dfrac{f_0}{\Delta}$ *is a modular form of weight* $l - 12$, *then because*

$$\frac{1}{\Delta} = \frac{1}{q}\prod_{i=1}^{+\infty}\frac{1}{(1-q^i)^{24}} = \frac{1}{q}\prod_{i=1}^{+\infty}\left(\sum_{j=0}^{+\infty} q^{ij}\right)^{24}$$

then $\dfrac{f_0}{\Delta}$ *is also of integral coefficients, thus finishes the induction.*

7. Let $f \in M_k(1)$, show that $\theta_k(f) = \frac{1}{2\pi i}\frac{d}{dz}f - \frac{k}{12}E_2 \cdot f$ defines a linear map $\theta_k : M_k(1) \to M_{k+2}(1)$.

证明 (解答). *By the definition of* M_k, *for all* $\gamma = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z})$,

$$f(\gamma z) = (cz+d)^k f(z),$$

$$(cz+d)^{-2} f'(\gamma z) = ck(cz+d)^{k-1} f(z) + (cz+d)^k f'(z).$$

By the proposition used in hw2.3,

$$E_2(\gamma z) = (cz+d)^2 E_2(z) + \frac{6c(cz+d)}{\pi i}.$$

So we have:

$$(\theta_k(f))(\gamma z) = \frac{1}{2\pi i}f'(\gamma z) - \frac{k}{12}E_2(\gamma z)f(z) \tag{1.7}$$

$$= \frac{1}{2\pi i}[ck(cz+d)^{k+1}f(z) + (cz+d)^{k+2}f'(z)] - \frac{k}{12}[(cz+d)^2 E_2(z) + \frac{6c(cz+d)}{\pi i}](cz+d)^k f(z) \tag{1.8}$$

$$= \frac{1}{2\pi i}(cz+d)^{k+2}f'(z) - \frac{k}{12}(cz+d)^{k+2}E_2(z)f(z) \tag{1.9}$$

$$= (cz+d)^{k+2}(\theta_k(f))(z). \tag{1.10}$$

8. Let L be a lattice, show that $\mathbb{C}/L$ is a compact Riemann surface of genus 1.

证明 (解答).

- *Let* $L = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2$ *be the lattice, where* $\omega_1, \omega_2 \in \mathbb{C}$ *are linearly independent on* $\mathbb{R}$, *then the closed parallelogram* $P = \{a\omega_1 + b\omega_2 :\ a, b \in [0,1]\}$ *induces a projection* $P \to \mathbb{C}/L$, *thus* L *is compact. Hausdorff property and connectedness come easily from the property of* P.

  For each $p \in \mathbb{C}/L$, *choose* $z \in \mathbb{C}$ *with* $\pi(z) = p$. $\pi|_{D(z,\varepsilon)} : D(z,\varepsilon) \to U := \pi(D(z,\varepsilon))$ *is a homeomorphism. Define a chart* (U, φ), *where* $\varphi = (\pi|_{D(z,\varepsilon)})^{-1}$.

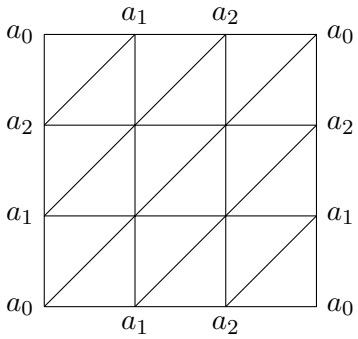
  Suppose (U_1, φ_1) *and* (U_2, φ_2) *are two such charts with* $U_1 \cap U_2 \neq \emptyset$. *Let* $V_i = \varphi_i(U_i)$. *On a connected component of* $\varphi_1(U_1 \cap U_2)$, *the composition* $\varphi_2 \circ \varphi_1^{-1}$ *satisfies*

$$\pi(\varphi_1^{-1}(z)) = \pi(\varphi_2^{-1}((\varphi_2 \circ \varphi_1^{-1})(z))),$$

so $\varphi_1^{-1}(z)-\varphi_2^{-1}((\varphi_2\circ\varphi_1^{-1})(z)) \in L$. *Since both points lie in small disks and* L *is discrete, this difference is constant on the component. Thus,* $\varphi_2 \circ \varphi_1^{-1}$ *is a translation by a lattice element, hence holomorphic.*

These charts form a holomorphic atlas, making $\mathbb{C}/L$ *a Riemann surface.*

- *Secondly, we check it is of genus* 1.



Because $\mathbb{C}/L \cong P/\{b\omega_2 \sim \omega_1 + b\omega_2, a\omega_1 \sim a\omega_1 + \omega_2\}$, *we know* $\mathbb{C}/L$ *is homeomorphic to* T^2,

By the triangulation of T^2 *above, we have* $2 - 2g(T^2) = V + F - E = 9 + 18 - 27 = 0 \Rightarrow g(T^2) = 1$.

9. Let $\mathbb{H}^* = \mathbb{H} \cup \{\infty\}$, and $\pi : \mathbb{H}^* \to X(1)$ be the projection, let $D_0 = \frac{1}{2}\pi(i) + \frac{2}{3}\pi(\rho) + \pi(\infty) \in \mathrm{Div}_{\mathbb{Q}}(X(1))$ be a $\mathbb{Q}$-Cartier Divisor, prove that the map:

$$M_{2k}(1) \to \{\omega \in \Omega^k_{X(1)} : \mathrm{div}(\omega) \geq -kD_0\} \tag{1.11}$$

$$f \mapsto \omega_f = f(z)(dz)^k \tag{1.12}$$

gives an isomorphism, conclude from it that $M_{2k} \cong \mathfrak{O}(kK_{X_1} + [kD_0])$, then use Riemann-Roch Theory to calculate $\dim M_{2k}(1)$.

证明 (解答). *We give a proof of the generalized case (where* $\mathrm{SL}_2(\mathbb{Z})$ *may be replaced by any congruence group* Γ*).*

- *First we know that weight-*k *modular form can be realized as a* $\frac{k}{2}$*-holomorphic differential on the complete modular curve* $X(1)$*, so the map* $M_{2k}(1) \to \Omega^k_{X(1)}$ *is well-defined. We want to prove that there is an isomorphism:* $A_{2k}(\Gamma) \to \Omega^k_{X(\Gamma)}$*, where* A_{2k} *denotes the modular functions (only meromorphic needed).*

  The realization is given as follow:

  For $\omega \in \Omega^k_{X(\Gamma)}$*, directly define the pullback of the natural projection*

$$\pi : \mathbb{H} \to X(\Gamma), \quad \pi^*(\omega) \in \Omega^k_{\mathbb{H}},$$

because $\mathbb{H}$ *is an open subset of* $\mathbb{C}$, $\pi^*(\omega)$ *can be written as* $f(z)(dz)^k$*, where* f *is a global meromorphic function, and the transition compatibility*

$$g^*(f(z)(dz)^k) = f(z)(dz)^k, \forall g \in \Gamma$$

gives:

$$f(gz) = (cz+d)^{-2k} f(z),$$

which means f is modular of weight-$2k$.

For $f \in A_{2k}(\Gamma)$, we will construct a meromorphic differential $\omega(f) \in \Omega^k_{X(\Gamma)}$ such that $\pi^*(\omega) = f(\tau)(d\tau)^k$. By the definition of meromorphic differentials it suffices to construct local differentials that pull back to restrictions of $f(\tau)(d\tau)^k$. Thus the idea is to express $f(\tau)(d\tau)^k$ in local coordinates.

Each local map $\psi_j : U_j \longrightarrow V_j$ is a composite $\psi_j = \rho_j \circ \delta_j : \tau \mapsto z \mapsto q$ with $\delta_j \in \mathrm{GL}_2(\mathbb{C})$. Since δ_j is invertible it is easy to transform $f(\tau)(d\tau)^k$ into z-coordinates locally. First extend the weight-k operator $[\gamma]_k$ to matrices $\gamma \in \mathrm{GL}_2(\mathbb{C})$ by defining $j(\gamma,\tau) = c\tau + d$ as we define on the $\mathrm{SL}_2(\mathbb{Z})$, and

$$(f[\gamma]_k)(\tau) = (\det\gamma)^{k/2} j(\gamma,\tau)^{-k} f(\gamma(\tau))$$

when this makes sense. Now let $U_j' = U_j \cap \mathbb{H}$. Then $f(\tau)(d\tau)^k|_{U_j'}$ is a pullback $\delta_j^*(\lambda_j)$ where λ_j is the differential obtained by pulling $f(\tau)(d\tau)^k|_{U_j'}$ forward to z-space under the change of multiplication by δ_j^{-1}. Letting $\alpha = \delta_j^{-1}$,

$$\begin{aligned}\lambda_j &= \alpha^*(f(\tau)(d\tau)^k|_{U_j'}) = f(\alpha(z))(d(\alpha(z)))^k \\ &= (\det\alpha)^k j(\alpha,z)^{-2k} f(\alpha(z))(dz)^k = (f[\alpha]_{2k})(z)(dz)^k.\end{aligned}$$

Thus λ_j is $\delta_j\Gamma\delta_j^{-1}$-invariant since $f(\tau)(d\tau)^k$ is Γ-invariant.

Denote r_h the multiplication by $\mu_h = e^{\frac{2\pi i}{h}}$.

When U_j does not contain a cusp, by $\delta_j\Gamma\delta_j^{-1}$-invariance, the pullback

$$r_h^*(\lambda_j) = (f[\alpha]_{2k})(\mu_h z)\cdot \mu_h^k (dz)^k = \lambda_j,$$

Or equivalently,

$$(\mu_h z)^k (f[\alpha]_{2k})(\mu_h z) = z^k (f[\alpha]_{2k})(z),$$

This shows that the function $z^l(f[\alpha]_{2k})(z)$ takes the form $g_j(z^h)$ for some meromorphic function g, and then we define:

$$\omega_j = \frac{g_j(q)}{(hq)^k}(dq)^k \ \text{ on } V_j. \tag{1.13}$$

When U_j contains a cusp, we similarly have the conclusion that $f([\alpha]_{2k})(z)$ takes the form $g_j(q_h)$, where h is the width and $q_h = e^{\frac{2\pi i z}{h}}$, then we may define:

$$\omega_j = \frac{g_j(q)}{(2\pi i q/h)^k}(dq)^k \ \text{ on } V_j. \tag{1.14}$$

This finishes the construction of $\omega = \{\omega_j\}_{j\in J}$. By tedious calculation we may see these two maps gives out a bijection.

Let τ_j be the center of U_j, from (1.13) *and* (1.14) *we know*

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\begin{align}
   v_{\pi(\tau_j)}(\omega) &= v_0(\frac{g_j(q)}{q^k}) \\
   &= v_0(g_j) - k \\
   &= \frac{1}{h}(v_{\tau_j}(f)) + \frac{n}{h} - k \\
    &= v_{\pi(\tau_j)
```

1.3　试卷: 模形式

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You may use all the propositions that have been proved in the exercises directly.

Some q-expansions you may use:

$$E_4(z) = 1 + 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n$$

$$E_6(z) = 1 - 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n$$

$$E_{14}(z) = 1 - 24 \sum_{n=1}^{\infty} \sigma_{13}(n) q^n$$

$$\Delta = q - 24q^2 + \dots$$

1. (1) Let E_k be the weight-k normalized Eisenstein series. Prove that $E_4^2 E_6 = E_{14}$.
   (2) For $f \in M_k(\Gamma(1))$, $g \in M_l(\Gamma(1))$, prove that $kfg' - lf'g \in M_{k+l+2}(\Gamma(1))$.
   (3) Prove that $j' = -2\pi i \frac{E_{14}}{\Delta}$, where $j = \frac{E_4^3}{\Delta}$.
2. Let $K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O}_K = \mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$.
   (1) Find a lattice $\Lambda \subseteq \mathbb{C}$, such that $E(\Lambda)$ has complex multiplication by $\mathcal{O}_K$.
   (2) Find $\mathrm{Aut}(E(\Lambda))$.
   (3) Use this elliptic curve to find $\mathrm{Cl}(K)$.
   (4) Find an elliptic curve defined over $\mathbb{Q}$ which is isomorphic to $E(\Lambda)$.
3. Solve the following moduli question: For a prime p, prove that there is a one to one correspondence between $S_0(p) = \{[E, C], C \subseteq E[p]$ is a cyclic subgroup of order $p\}/\sim$ and $Y_0(p) = \Gamma_0(p)\backslash\mathbb{H}$.
4. (1) Prove that $M_k(\Gamma_0(N)) = 0$ when k is odd.
   (2) Let p be a prime, $S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Prove that

$$\Gamma(1) = \Gamma_0(p) \bigsqcup \bigsqcup_{k=0}^{p-1} \Gamma_0(p) S T^k.$$

   (3) Prove that $M_2(\Gamma_0(2))$ is one-dimensional.